Lecture Transcript by Slide

Week 6 · Chapter 4: Sinus Rhythms · 55 min

Auto-transcribed, so expect occasional errors, especially in technical terms. Timestamps are positions in the recording.

0:00:00 Chapter 4: Sinus Rhythms (title slide)

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0:19:26 Sinus Bradycardia (ECG strip)

0:33:43 Mechanism of Sinus Tachycardia

0:34:10 Rules for Sinus Tachycardia

0:35:27 Sinus Tachycardia (ECG strip)

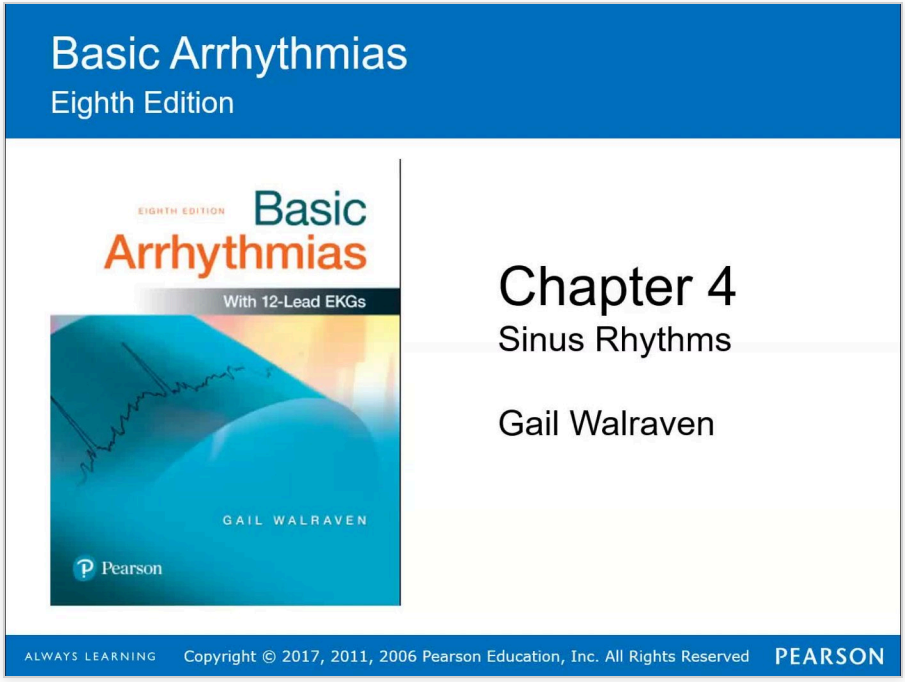
0:42:29 Mechanism of Sinus Arrhythmia

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1 Chapter 4: Sinus Rhythms (title slide)

0:00:00 – 0:01:15

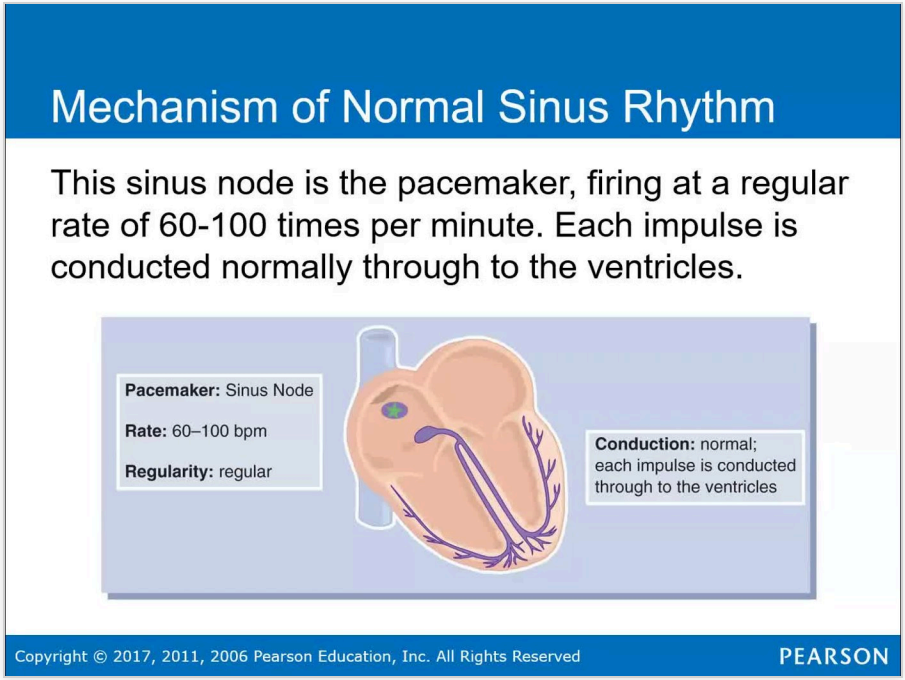


0:00:24 Michael V Hyland: Okay. So now we're gonna move into actual rhythm interpretation. Michael V Hyland: So we're gonna start today with Michael V Hyland: sinus rhythms. So there's 4 different rhythms Michael V Hyland: that fall in this category. When we say sinus rhythms, it means all of these rhythms originate from the sa node.

0:00:52 Michael V Hyland: So there's 4 different rhythms that we're gonna cover. In this lecture Michael V Hyland: you have a homework assignment with 10 strips on it based off of this lecture. Michael V Hyland: So all of the strips on that homework assignment will be one of the 4 rhythms that we cover in class today. Michael V Hyland: Okay. Michael V Hyland: alright. So the first one I'm going to start with is normal sinus rhythm.

2 Mechanism of Normal Sinus Rhythm

0:01:15 – 0:03:20



0:01:23 Michael V Hyland: So Michael V Hyland: the essay node in this is the primary pacemaker. Michael V Hyland: It's going to fire at regular intervals

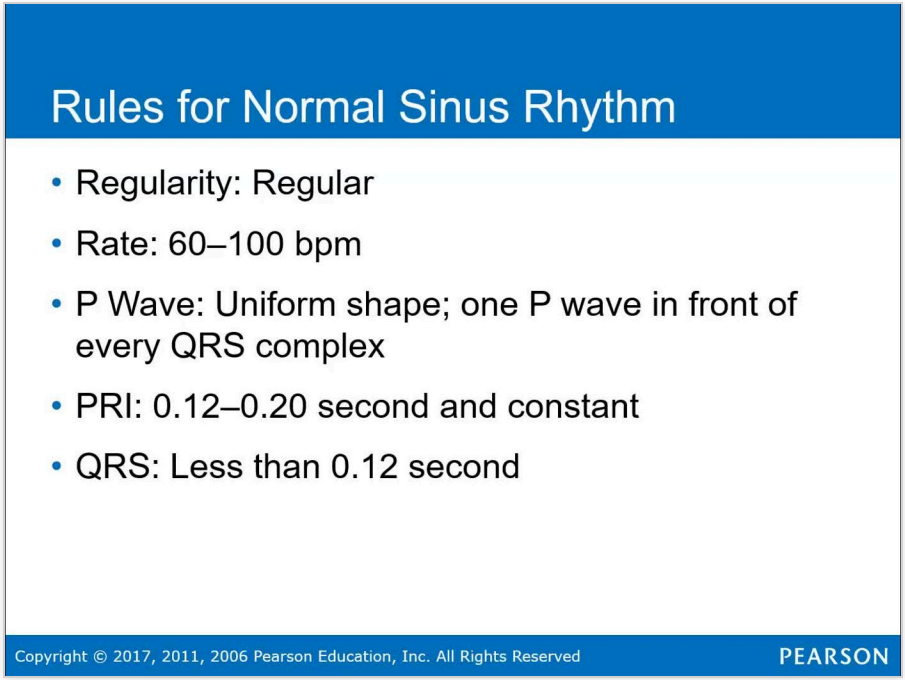
0:01:39 Michael V Hyland: a rate between 60 to 100. So when I say, fire at regular intervals. What does that automatically mean? Michael V Hyland: That automatically means our to our intervals going to be regular? So when I say regular intervals, that means that our waves will be evenly spaced. Michael V Hyland: Okay, so I'm just trying to help you out with some of the terminology. The way that the publisher prints the slides right? Firing at a regular rate of 60 to 100 and regular intervals. Okay? Michael V Hyland: And then Michael V Hyland: each impulse is conducted normally through the ventricle ventricles. Michael V Hyland: So what does that mean? That means that the electrical impulse travels through the A B node through the bundle of his. The right and left bundle branches is delivered to the brackenji fibres.

0:02:33 Michael V Hyland: So when conduction is normal, that means it's going along the nerve impulse pathway. and that means the electrical impulse is delivered quickly. Michael V Hyland: and ventricular contraction happens in a normal fashion. Michael V Hyland: What's a normal QRS.

0:02:53 Michael V Hyland: Point 1 2 s, or less, or 3 or small? 3 boxes or less right. So all of these Michael V Hyland: Q. Rss. Are going to be narrow. Q. Rss. Michael V Hyland: Normal or narrow. QR. S. Remember, 2 weeks ago I use those 2 words interchangeably normal or narrow. You need to understand that that is point 1 2 s or less. Michael V Hyland: Patty.

3 Rules for Normal Sinus Rhythm

0:03:20 – 0:05:35



0:03:20 Michael V Hyland: So we have a set of guidelines. So remember. way back in the first semester I talked about, we're gonna are in the the first and second lecture we talked about. We're going to gather clues. Michael V Hyland: We're gonna gather clues. And then we're gonna compare these clues Michael V Hyland: to Michael V Hyland: the guidelines for each rhythm to see where it best fits. And that's how we interpret. Well, here are the guidelines for normal sinus rhythm. Michael V Hyland: So the regularity is going to be regular. So this our spacing is going to be regular across the strip. The heart rate's going to be in the normal range of 60 to 100 beats per minute.

0:04:04 Michael V Hyland: P. Waves will all be upright Michael V Hyland: dome shape. So Sinus P. Waves are positively deflected or upright. and they usually have a nice dome shape to them.

0:04:20 Michael V Hyland: So Sinus P. Waves will be upright. They'll all look the same, so they'll be upright and uniform. or positively deflected, with a nice dome shape.

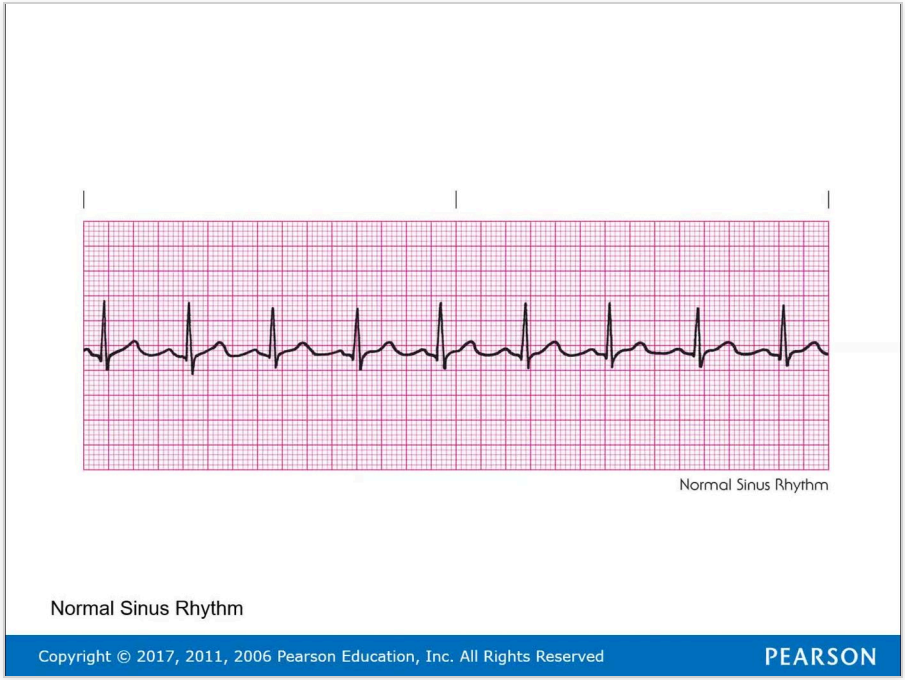
0:04:35 Michael V Hyland: They don't always have to have a nice dome shape, but they're typically not Michael V Hyland: pointed or flattened or notched. So there's usually a roundness to the top of the sinus P wave. Michael V Hyland: It's going to have a normal Pr, I, because we have we're gonna have. When the sa node fires the fire. The impulse goes to the intra atrial pathway, causing the atri to contract. That's why we get a P wave. Michael V Hyland: And then we're gonna have a small period of electrical inactivity after each of contraction. That's the delay at the Ab node right? And then we're going to send it down for ventricular contraction.

0:05:15 Michael V Hyland: So we're going to have a normal delay of 3 to 5 small boxes or point 1 2 to point 2 s. Michael V Hyland: and then the Qrs, I already said, is going to be normal or narrow.

0:05:29 Michael V Hyland: So that is going to be point 1 2 s or less so now, what I want to do here is throw up a rhythm, and you can see if you take a quick look at the homework assignment. We're still going to be assessing regularity

4 Normal Sinus Rhythm (ECG strip)

0:05:35 – 0:18:05



0:05:45 Michael V Hyland: heart rate. P. Waves. Pr. I. Michael V Hyland: Qrs. And now we're actually going to interpret each rhythm. Michael V Hyland: So if you look at that homework assignment. Each strip has 6 spots underneath it that you should be filling out for each rhythm. Michael V Hyland: and each one of those is going to be worth 1 point. Okay. so let's go through our information gathering on this strip. Michael V Hyland: I would expect that you would follow along and write it down on your Powerpoint, so that you have something to refer to when you go home and try to do the homework.

0:06:24 Michael V Hyland: Okay. Michael V Hyland: so what's our first step? Michael V Hyland: Pound the number of small boxes between our waves. Michael V Hyland: 510-15-1617.

0:06:47 Michael V Hyland: I think it's 17 small boxes between the first and the second R. Wave. 5, 10, 13, and 4 makes 17, 510-15-1617,

0:07:03 Michael V Hyland: 5, 1015, 17, Michael V Hyland: 5, 13, and 4 make 17, Michael V Hyland: 17, Michael V Hyland: 5, 10, Michael V Hyland: 1415, 1617, I think, between Michael V Hyland: these last 2, not the last 2, the second to the last 2. There's 17 and a half small boxes that our wave is right in the middle.

0:07:34 Michael V Hyland: 510-15-1617, and a half. So we have 1717, 1717, 1717 17 and a half 17 and a half. Michael V Hyland: so everybody understand how I got that?

0:07:52 Michael V Hyland: Okay. Michael V Hyland: so what's my heart rate, is it? I'm sorry. Is this rhythm regular or irregular? Michael V Hyland: What's the cutoff between regular and irregular Michael V Hyland: one and a half small boxes, right? So all of these spaces are within one and a half small boxes of each other. So that is gonna deem this rhythm regular.

0:08:15 Michael V Hyland: How do we count the heart rate when the rhythm is regular? Diane? Thank you. So number of small boxes divided into 1,500.

0:08:27 Michael V Hyland: Our rates gonna be 88. Okay. Michael V Hyland: so regular, with a heart rate of 88 pea waves does this rhythm have P waves? Michael V Hyland: It does.

0:08:46 Michael V Hyland: Are they positively deflected they are. Michael V Hyland: Do they? Pretty much look the same? Michael V Hyland: I would agree, I think, as you, as you look at the shape of each one of these P. Waves that they're pretty close like. If you overlaid one over the other they would look Michael V Hyland: very similar. So P. Waves are

0:09:12 Michael V Hyland: uniform and upright. I like it. What's the next thing we do? Michael V Hyland: Measure the P. Ri. How do we measure the PRI measure from where to where. Thank you. Michael V Hyland: Beginning of the P. Wave, the beginning of the QRS.

0:09:31 Michael V Hyland: So if you look at some of these P. Waves, it gets a little fuzzy. That upslope is kind of slanted on some of them Michael V Hyland: right

0:09:42 Michael V Hyland: like the third P wave in right? That's a little slanted there. Michael V Hyland: so I'm probably not going to. I mean, I might look at that one, but it's not going to be one of the primary ones that I would look at. So we got to find the beginning of the P wave. This first one

0:10:00 Michael V Hyland: looks to be about right. There is the start of the upslope. right at the beginning of that box where my cursor is

0:10:10 Michael V Hyland: right there. and the beginning of the see the negative drop right there. Michael V Hyland: So that looks like one complete large box. So you start at the beginning of the negative drop. Michael V Hyland: Yes. beginning of the Qrs. So if we look at this virtual, the upsold starts right there. Michael V Hyland: and the Q-way. Look at the negative. Qa starts right there.

0:10:38 Michael V Hyland: You guys agree with that pointed out over here as well. Michael V Hyland: So on this first P wave the up slope. P-wave starts right there. Michael V Hyland: The downsope, the native Qwa starts right there Michael V Hyland: so that first one would be 5. Right? So we need to count a couple of more to make sure that Michael V Hyland: the pri is consistent across the strip Michael V Hyland: can't just count one, because eventually we're going to do rhythms where the Pri is going to change.

0:11:18 Michael V Hyland: So I think the second Pr. I is also 5 small boxes. So if you see where my cursor is, it starts right there.

0:11:27 Michael V Hyland: and it ends right there. There's a little negative drop there. So that is 1, 2, 3, 4, 5. Michael V Hyland: Want to skip that one there. Michael V Hyland: So I'm going to go here, 1, 2, 3, 4,

0:11:48 Michael V Hyland: 5, that one might just be just short of 5 boxes. So if you say, if I say like, it's 4 and a half boxes like. Michael V Hyland: So maybe you take it. Michael V Hyland: So remember, last week, I said. You know, even on the test that you handed in today and the homework assignment. I give you one small box, Grace, so as long as your answer is within one small box of my answer, I won't mark it wrong. So if you put 4 and a half. Michael V Hyland: how much time is 4 and a half. By the way.

0:12:18 Michael V Hyland: times 4.5 Michael V Hyland: point 1 8 4 and a half small boxes would be point 1 8. I'm saying it's once one large box or 5 small boxes, so I would have point 2. You would have point Michael V Hyland: that that's within one small box. I'm not gonna mark it wrong. Alright.

0:12:40 Michael V Hyland: I'm going to do one more over on this side Michael V Hyland: 1, 2, Michael V Hyland: 3, 4, and that one is 4 and a half. So the last 2 that I've counted is 4 and a half.

0:12:54 Michael V Hyland: This next one is 5. Michael V Hyland: This next one is 5. So listen. Michael V Hyland: we're eyeballing this the best we can. This is not and exact science. When you do a 12 lead. Ekg, the Ekg machine actually measures it for you. Michael V Hyland: We don't have that luxury. You guys need to know how to measure the Pri Michael V Hyland: as the pri is one of the clues that's going to help us identify where the rhythm is coming from. Michael V Hyland: Okay. So Michael V Hyland: you know, as long as you're in the ballpark, I think you'll be okay, right?

0:13:36 Michael V Hyland: And remember the tricks that I showed you. If you put a piece of paper on there and put a mark on the piece of paper where it starts. Remember, I showed you this last week. Michael V Hyland: Right? Put a mark where it starts, put a mark where it where it ends. Michael V Hyland: and then bring it down in the box sometimes easier to see it that way.

0:13:59 Michael V Hyland: I'm pretty confident this pri is 5 small boxes. Okay. Michael V Hyland: what do we measure next? So we're going to measure the QRS. So let's go ahead and measure that.

0:14:23 Michael V Hyland: I actually think it is 2 and a half small boxes. Michael V Hyland: You want to know how I derive that.

0:14:35 Michael V Hyland: Okay. let's look at the first one. Michael V Hyland: I'm gonna look at this first one. We said that the negative Q wave starts right on the dark line. So that's where the Qrs starts. Beginning of the Q wave. Look where the S. Wave ends right here in the middle of these 2 boxes. Michael V Hyland: so if you count Michael V Hyland: over one. Michael V Hyland: 2, and a half.

0:15:03 Michael V Hyland: I'll point it out. Up. Here. Remove the cursor. I'll point it out up here. I'll do it on both sides. Michael V Hyland: So the negative, anyway, starts right there right on this line. Michael V Hyland: Thank you very much. Whoever this patient was, make it easy for me to count. Start right out of line, right very thoughtful of them. Michael V Hyland: And then this is where the S wave ends. Right there in the middle of those 2 boxes.

0:15:28 Michael V Hyland: 1, 2 and a half. Michael V Hyland: Everybody see how I got that Michael V Hyland: same thing. Guys over here this first. Yeah, URL starts right on that line Michael V Hyland: right there and the best way. Then look at it's halfway in between these 2, Michael V Hyland: 1, 2, and a half. Michael V Hyland: So 2 and a half small boxes makes up how much iron

0:15:59 Michael V Hyland: point? Oh, 8 plus point oh, 8 plus point oh, 2 point 1 0. So the Qrs here is point 1 0 Michael V Hyland: alright. So I'm gathered all my clues. The rhythm is regular, my heart rate is 88 heat waves are positive, they all look the same. They're nice and rounded. Pri is normal. Qrs is normal. This rhythm because everything is normal. Michael V Hyland: Only one place this is coming from. This is normal science. Rhythm would be my interpretation. Here Michael V Hyland: everybody follow. You have a question.

0:16:35 Michael V Hyland: Don't be shy. Michael V Hyland: So what did you grow when you need to vote for me about? Michael V Hyland: I would rather have the time measurement. Michael V Hyland: It would

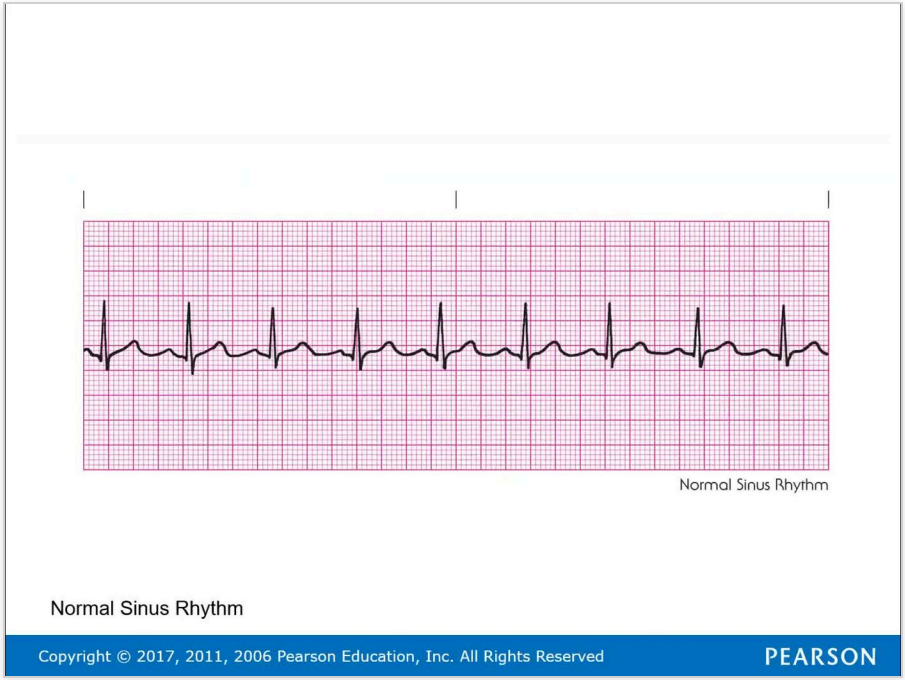
0:16:52 Michael V Hyland: probably not on the homework. But when it comes to the test. Michael V Hyland: So here's the thing is on the next test. So the next test Michael V Hyland: is not going to be a multiple choice quest. The next test is going to look just like your homework assignments. There will be 15 strips on that next test. Michael V Hyland: By the time we get to that next test we will have covered sinus rhythms, atrial rhythms and junctional rhythms. So all 3 of those rhythm sets will be on that 15 strip test. Michael V Hyland: and then the next test after that will be the same thing, except that will be ventricular rhythms, heart block rhythms and paste rhythms. But the final exam is back to multiple choice.

0:17:37 Michael V Hyland: And I ask you to measure pris and Qrs is on that final exam, and they are. The answers are in time, not in boxes. Michael V Hyland: so I think you should try to. You know. Michael V Hyland: Mass it up a little bit scope. Michael V Hyland: You can do it. I

don't. I know you can alright questions on how we got here.

5 Mechanism of Sinus Bradycardia

0:18:05 – 0:18:53



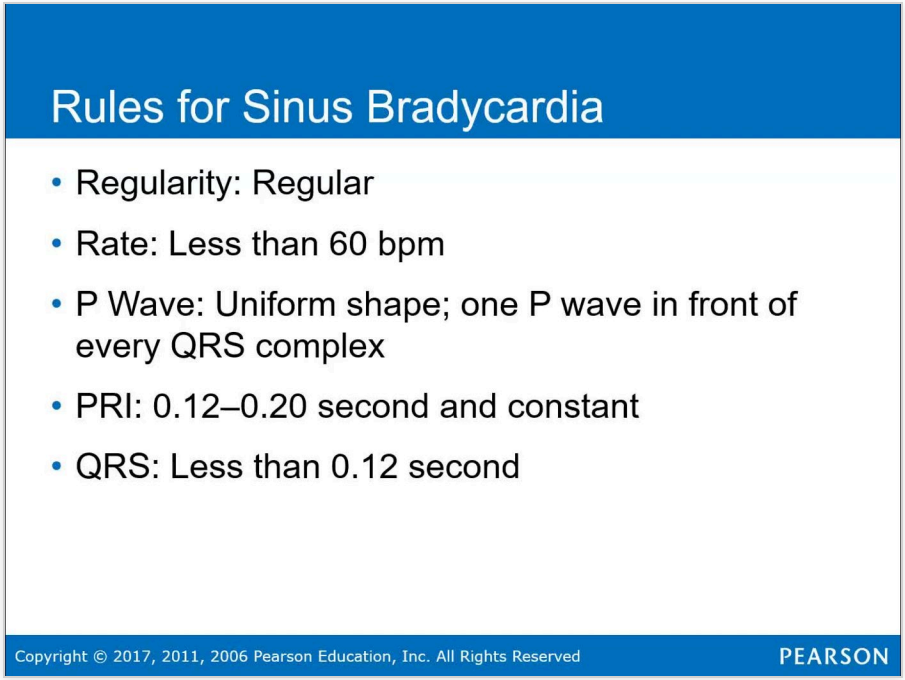
0:18:05 Michael V Hyland: If we again, if we go back and compare it to the rules here, everything fits Michael V Hyland: right in that rule. Okay. Michael V Hyland: all right, let's move on to the next one. So now we're gonna talk about Sinus Brad, a cardiac. What does the word bradycardia mean?

0:18:22 Michael V Hyland: Heart rate? Less than 60? Okay. Michael V Hyland: so this rhythm also comes from the sa node. It still fires at regular intervals. except now it just fires at a rate of less than 60.

0:18:42 Michael V Hyland: Each impulse is conducted normally through the ventricle. So that means I'm going to have a narrow or normal. QRS, okay. so let's look at the guidelines. This rhythm is regular. That means R to R is evenly spaced.

6 Rules for Sinus Bradycardia

0:18:53 – 0:19:26

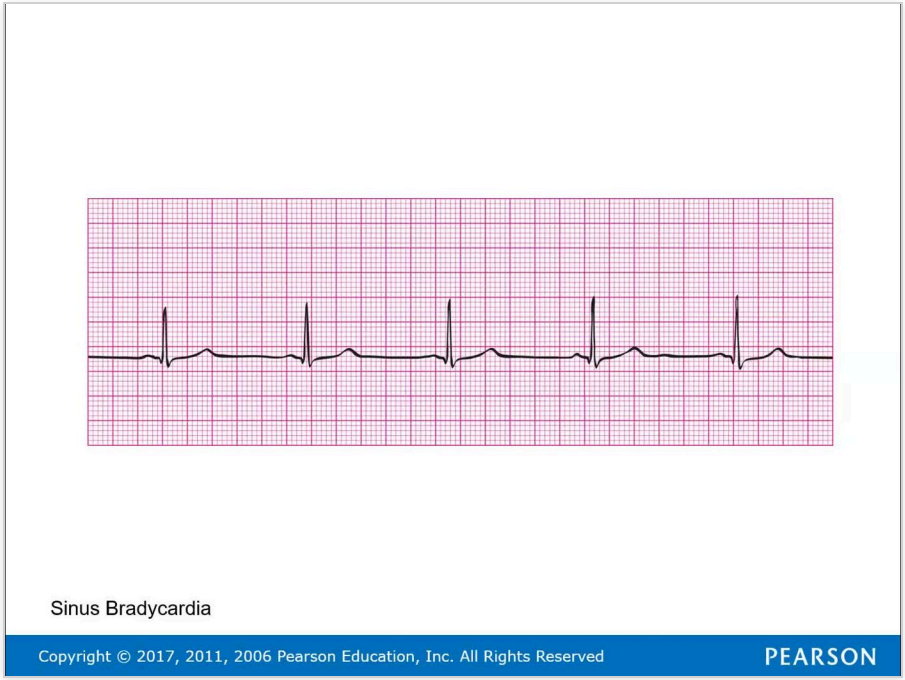


0:18:59 Michael V Hyland: Heart rate's going to be less than 60. So that means the R. Waves are going to be farther apart. right? Because that means I'm going to be dividing a larger number into 1,500 to get me that lower rates

0:19:14 Michael V Hyland: P. Waves will be upright in uniform. There'll be Sinus P. Waves, the Pr. I will be normal, and the Qrs will be narrow.

7 Sinus Bradycardia (ECG strip)

0:19:26 – 0:33:43



0:19:26 Michael V Hyland: So let's take a look at this. Anytime the rate is slower. Michael V Hyland: Our waves are farther apart. Any time the rate is faster. Our waves will be closer together. Michael V Hyland: So let's go through our information gathering here.

0:19:42 Michael V Hyland: Why don't you guys try to do it yourself? Count the number of our small boxes between our ways. Michael V Hyland: Are you guys good? Or what do we get for the first basing? Michael V Hyland: 28 and a half? Michael V Hyland: What do we get for the second, spacing. 29. When we get to the third spacing? 29. What do we get for the last spacing? 28 and a half?

0:21:17 Michael V Hyland: That's close to 29. It's leaning towards that line. Michael V Hyland: Sorry. Michael V Hyland: What's that triple Vietnam? Michael V Hyland: Well, we know who's gonna win. Michael V Hyland: There'll be no argument anyways. Michael V Hyland: And I'm not sure that it really matters, because if you, if you divided Michael V Hyland: 28.5 in the 1,500, somebody do that math for me

0:21:50 Michael V Hyland: 52.8 versus 1,500, divided by 29. You want to. Michael V Hyland: man, that's a big difference. Now I will tell you. I will tell you if you're off by a half a box or a box when their heart rate is this slow? You're gonna be pretty close to my answer. Michael V Hyland: But you make that mistake on a rhythm that's fast above a hundred. You could be 20

beats off 30 beats off. So you know, as we move on in the semester. You need to get more and more comfortable with counting the number of small boxes. Michael V Hyland: Okay. so this heart rate is 51, 52,

0:22:31 Michael V Hyland: pri, would you guys go ahead and measure the Pri. Michael V Hyland: Alright, what do you think? What'd you guys get?

0:23:38 Michael V Hyland: So I got a 4. I got a 3 and a half. Michael V Hyland: so I think one or 2 of these is close to 3 small boxes, but I think in general this is 3 and a half small boxes.

0:23:54 Michael V Hyland: Are you confused at how I got back? Michael V Hyland: This is the time to be honest with me. If you don't know how to do it. Michael V Hyland: I want to make sure you know how to do it before we move on. So you're comfortable over there. Michael V Hyland: forward to Michael V Hyland: Matthew. Okay. Michael V Hyland: well, it's within a half a box of mine, so I wouldn't mark it wrong. Michael V Hyland: Right? Remember, you're you're you're. Michael V Hyland: you know, on some of these P. Waves. I will be honest on some of these P. Waves. It's a little bit of a guess as to where it starts when the P. Waves Michael V Hyland: slopes up like that last P wave. Where does it actually start?

0:24:33 Michael V Hyland: It's hard, right? But this one is very clearly defined right here. Michael V Hyland: The second to the last one is very clearly defined. 1, 2, 3, and I think that's one of the ones that's actually closer to 3 boxes as opposed to 3 and a half. Michael V Hyland: So like, I said, if I said 3 and a half, and you said 4. You're right. If I said 3 and a half, and you said Michael V Hyland: 3, you're still not gonna get it wrong? Right? You're still gonna be right. And again, it's not an exact science. I'm just trying to get you to be as close to that as you can. Okay.

0:25:10 Michael V Hyland: all right, let's measure the QRS. Michael V Hyland: What'd you get over here? What'd you get for that? QRS, Michael V Hyland: yeah. What'd you get? Parents. 2 and a half. Michael V Hyland: What'd you guys get all of them?

0:26:08 Michael V Hyland: That's not the point. Michael V Hyland: So you got one and a half Michael V Hyland: choices. Michael V Hyland: You got 2 yeah. 2 and a half Michael V Hyland: great. Got a pretty wide range here. one and a half.

0:26:42 Michael V Hyland: I really think. like one of the the easiest 2 to measure are the third and the fourth one that are on this strip. So if we go to the fourth one. Michael V Hyland: the negative drop starts halfway in between everybody. See that starts it starts right there in between.

0:27:04 Michael V Hyland: See that negative drop right there. It starts just before the pink line. Michael V Hyland: and then we have QRS. And then, as we come back up. Michael V Hyland: it gets back up right there. See where the cursor is. Michael V Hyland: So if I count from half

way one. 2, 3, Michael V Hyland: so I think this Qrs is 3 small boxes.

0:27:35 Michael V Hyland: Yep, am I smoking crack Michael V Hyland: most of the time we should be measuring it when it gets measuring the width until it gets back up to the Iso electric line. Okay, that's what I'm doing. So if you look at this one, it starts there and it comes up there. 1, 2, 3, and if I look at this one it's yours. Michael V Hyland: You guys see what I just did there. Michael V Hyland: alright. So you take down you take it. you are in that box on that one, and you're a little bit further away here. Yeah. Michael V Hyland: here is where it gets back to the Iso electric mine here. So not in the old cage. So from here.

0:28:43 Michael V Hyland: what's it there? Michael V Hyland: I know you're gonna bring you up here. You see where I was measuring that from. Okay, I gotta get back down. You're out. Michael V Hyland: That's it. Michael V Hyland: Okay? Michael V Hyland: And then, apparently, I skipped over the P. Waves. But tell me about these P. Waves. Okay. Michael V Hyland: they're upright, and they all look the same. Right? So Michael V Hyland: would that make it up

0:29:39 Michael V Hyland: so? Michael V Hyland: Yes, it would make it abnormal. We're not there yet when we meet in 2 weeks, when we meet in 2 weeks.

0:29:52 Michael V Hyland: the P. Waves that are different are the abnormal beats. We might have a regular underlying rhythm. But the P. Waves that are different would be abnormal. Michael V Hyland: Okay.

0:30:05 Michael V Hyland: so we have a regular rhythm with a heart rate of 51 P. Waves that are upright and uniform. Michael V Hyland: a normal PRI. And a normal QRS.

0:30:17 Michael V Hyland: So this is coming from the sinus rhythm, but because the rate is below 60. Now, our interpretation is sinus braidocardia, not normal sinus rhythm Michael V Hyland: are very clear on that. I'd like to talk about clinical implications of these rhythms.

0:30:38 Michael V Hyland: Remember, in the beginning of the semester I talked about Ekg's measure, electrical activity. They don't measure mechanical activity. Michael V Hyland: and in this case we could have normal electrical activity and normal mechanical activity. But this patient could still be somewhat symptomatic. Michael V Hyland: A heart rate of 52 is not normal. Now, mind you. Michael V Hyland: arresting or sleeping heart rate of 52 could certainly be normal. So if we measured somebody's heart rate heart rate when they're fast asleep. Michael V Hyland: 52 is normal. But when you're up moving around, having daily activity. 52 is not normal.

0:31:22 Michael V Hyland: There is a group where it would be normal during daily activity, and that would be athletes that are in really good shape. Marathon runners probably have a resting heart rate of 50 Michael V Hyland: 52 right? However, if somebody's heart rate is normally 90 or

even 85, 80. This heart rate of 52 is abnormal.

0:31:45 Michael V Hyland: So what might be the problem now? All of a sudden, if their heart rate dropped to 52 Michael V Hyland: in the very first week we talked about cardiac output. What's the formula for cardiac output Michael V Hyland: stroke volume times heart rate. So we probably don't have a problem with stroke volume here. but we might have a problem with heart rate. Right? So stroke, volume times, heart rate equals cardiac output. Michael V Hyland: Cardiac output is the only supplier to blood pressure. Michael V Hyland: So if your cardiac output dips.

0:32:23 Michael V Hyland: you might have a drop in blood pressure here, right? So what we would need to do is make sure. We assess this patient to make sure that they still have a palpable pulse and measure their blood pressure Michael V Hyland: right, and if their blood pressure drops significantly.

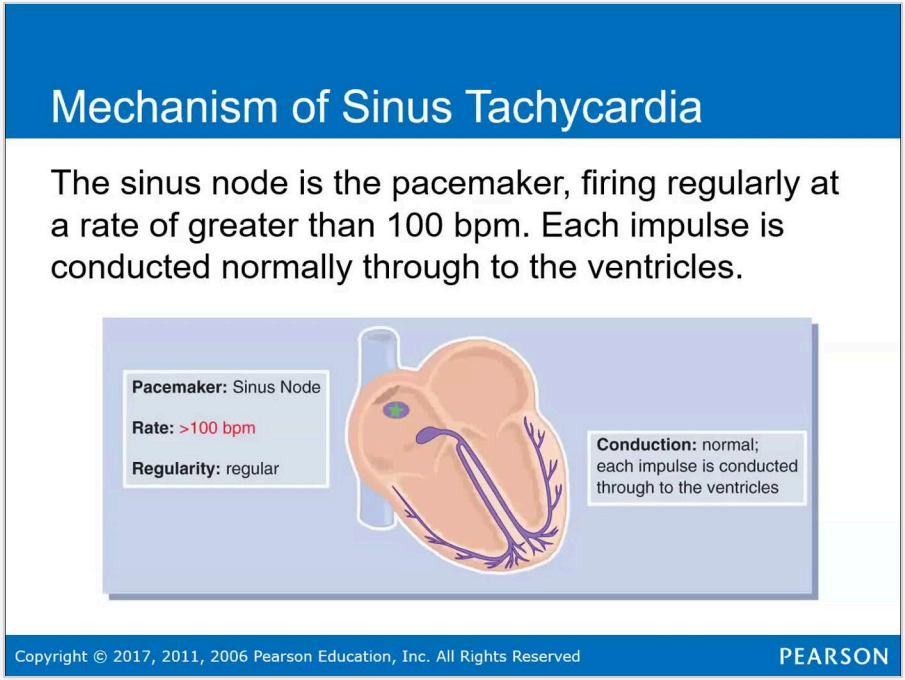
0:32:44 Michael V Hyland: then this rhythm could be potentially life, threatening, maybe not at a heart rate of 52. But if this heart rate drops down in the 40 s. Or even in the 30 s. It is definitely life threatening. Michael V Hyland: So if you're in a sleep lab somewhere, and you see somebody's heart rate drop down in the 40 s. Or 30 s. You better be picking up the phone and dialing 9 1 one, because that's abnormal right? Michael V Hyland: Especially if you assess them. And as your blood pressure drops, what happens to your mental state? Michael V Hyland: Because when your blood pressure drops, you're not refusing your brain. So this patient might be agitated, confused. When you touch their skin right. If their blood pressure is low, they might be sweaty and diaphratic. Right?

0:33:24 Michael V Hyland: So a little bit of patient assessment in conjunction with what you're seeing on the Monitor sometimes goes a long way right?

0:33:33 Michael V Hyland: So I just want to make sure you're aware of the rhythms that we cover in this class that could potentially be life, threatening

8 Mechanism of Sinus Tachycardia

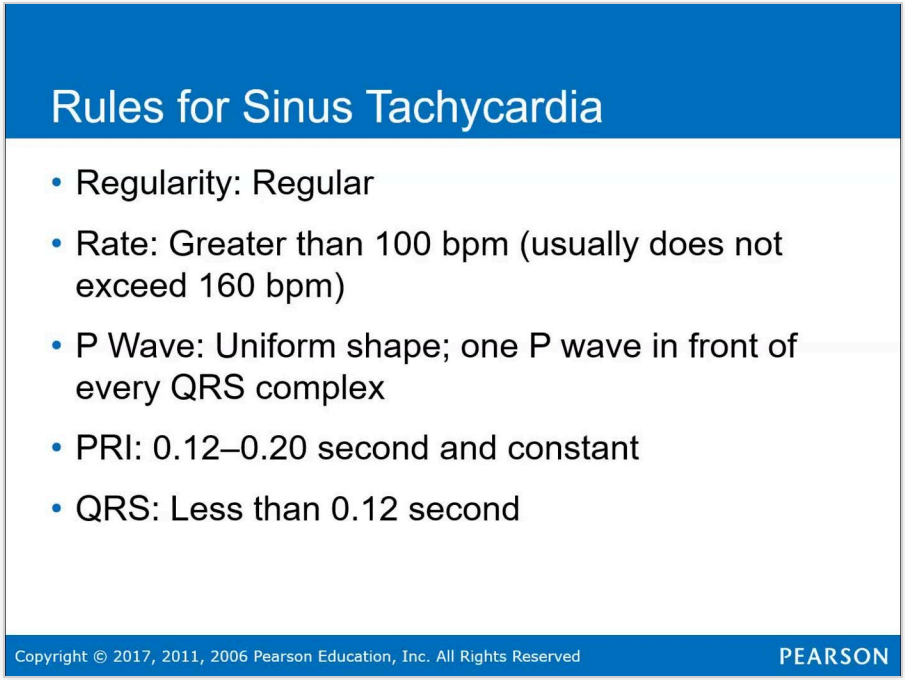
0:33:43 – 0:34:10



0:33:43 Michael V Hyland: all right onto the next rhythm. Sinus, tacha, cardia tacha cardia is a heart rate greater than 100. So again, the essay note is the pacemaker. Michael V Hyland: It fires at regular intervals. but now it fires at a rate greater than a hundred. and each impulse is conducted normally through the ventricle. So we have a regular interval Michael V Hyland: rhythm with a heart rate greater than a hundred, with narrow or normal. Q. Rss. so it's regular.

9 Rules for Sinus Tachycardia

0:34:10 – 0:35:27



0:34:14 Michael V Hyland: This heart rate's gonna be above a hundred. You don't need to know this fact today, but in the future you'll need to know this fact. Sinus tact Maxes out at a hundred 60. Michael V Hyland: So Sinus taco cardia in adults does not go higher than 1 60, it can go higher than that in kids, but not in adults. Okay.

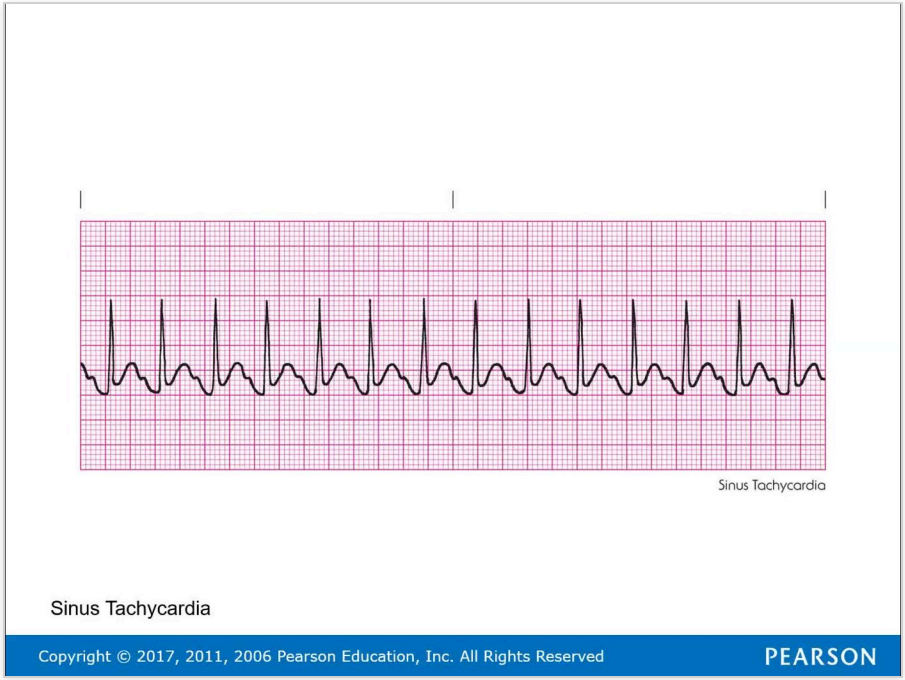
0:34:40 Michael V Hyland: P. Wave is going to be upright in uniform. and I want to add a caveat to that if we see it.

0:34:50 Michael V Hyland: Remember, we were talking about when the rhythm is slower, when the heart rate is slower, the R. Waves are farther apart. When the rhythm is faster the R. Waves are closer together. Michael V Hyland: and when the rhythm gets going fast enough in the high 100 fortys, 100 fiftys. There's not enough space in between complexes to actually see the P. Wave, or we might only see a partial aspect of the P. Wave.

0:35:20 Michael V Hyland: You'll see what I mean in a minute. PRI. Is going to be normal and the Qrs. Will be narrowed. So let's go through our information gathering.

10 Sinus Tachycardia (ECG strip)

0:35:27 – 0:42:29



0:35:32 Michael V Hyland: So count the number of spaces between our waves for me. Michael V Hyland: Yeah, I apologize. The strips that are on the Powerpoint. Michael V Hyland: I'm just. Here's the thing is, you know I've killed many forests in my day, so I'm trying not to kill another whole forest by printing single page slides

0:36:24 Michael V Hyland: so my apologies. I know I'm straight in your eyes. Michael V Hyland: any of any printing. Michael V Hyland: and then you

0:37:00 Michael V Hyland: I don't. Michael V Hyland: What do we come up with? Michael V Hyland: I would say 10 and a half. I don't see any 11 Si see maybe one or 2 that might Michael V Hyland: might be close to 10. But I think the majority of these are 10 and a half.

0:37:29 Michael V Hyland: Okay? So the rhythm's regular. Michael V Hyland: What's our heart rate up? Our spacing is 10 and a half. Michael V Hyland: I know what you did. Yeah, what is it? Michael V Hyland: 100 and 43.

0:37:57 Michael V Hyland: Okay. P. Waves. Michael V Hyland: So we can just barely see these P. Waves. Michael V Hyland: These P. Waves actually start on the down slope of the T. Wave. See that little notch on the bottom of the T wave right there. So that's all we got. So we're going to make our our P. Waves Michael V Hyland: upright, and they all pretty much look the same coming off of that bottom of the T wave

0:38:27 Michael V Hyland: so upright in uniform. And then the Pr. I. We're gonna count from the beginning of the notch to the beginning of the QRS.

0:38:56 Michael V Hyland: This one actually is probably a little bit easier to measure, I think. Michael V Hyland: as the beginning of the P wave is pretty clearly defined.

0:39:05 Michael V Hyland: and there's so here's the thing is, there's no Q wave in this one. So all you do is you measure to the beginning of the R wave Michael V Hyland: right? There's no negative deflection for the Qrs. There's no negative deflection right there. It just goes straight up. So you're measuring for the beginning of the p-wave, the beginning of the R ways. Michael V Hyland: the absolute of the R wave.

0:39:26 Michael V Hyland: It's no negative deflection there. Michael V Hyland: don't tell me. You see one. I do. A negative. Q. Wave. Michael V Hyland: Put down the bike

0:39:46 Michael V Hyland: alright. What do we got? Michael V Hyland: So we got 4. Michael V Hyland: Would you guys get over here, or

0:40:01 Michael V Hyland: I like 3 and a half. I think it's 3 and a half Michael V Hyland: alright. All right, let's measure the QRS.

0:40:11 Michael V Hyland: Again. This one's probably a little easier. Michael V Hyland: What'd you get? Or the Qrs. the excess. Michael V Hyland: I mean here. that's the beginning of the rs, okay. and there's the end of it right there. Then it goes right into the T-wave. So we're here. It's there.

0:40:53 Michael V Hyland: Did you count 2 or 2 and a half on every sign, yeah, keep. Yeah. And you're blocking everything. Michael V Hyland: My big head. It was like, you're like, 3 in the kitchen.

0:41:03 Michael V Hyland: Yeah. Rs, starts right there. Michael V Hyland: right R. Way down to the sway, and then it goes up into the T, so it ends right there. So from here to there. Oh, I was counting okay. Michael V Hyland: alright. So what do we got Michael V Hyland: for? Like 1, 2,

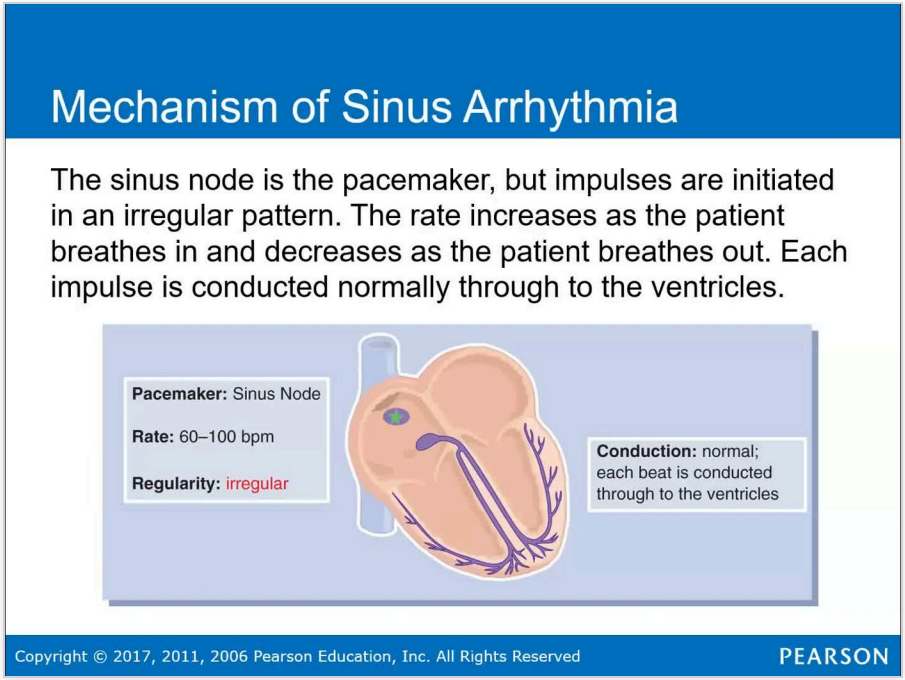
0:41:24 Michael V Hyland: I like one and a half. It's one and a half. But just show me that right it starts. Look at this and start right on the line again. Perfect. Thank you very much. Michael V Hyland: And then it ends halfway between right there. So I'm gonna count from here. Michael V Hyland: There. There's one small box, and then a half, a small box. Michael V Hyland: This is Michael V Hyland: becoming a little bit more clear.

0:41:53 Michael V Hyland: I love the confidence in that. That's awesome right there. Michael V Hyland: Remember, I told you not everyone is gonna have all the waves right? Not everyone's gonna have a Q wave, not everyone's gonna have an R wave. Not everyone's gonna have an S wave. Michael V Hyland: So all I'm asking is, you identify the beginning of the Qrs complex to the end of it and count the number of small boxes.

0:42:21 Michael V Hyland: See, we're making you very versatile here what we're doing. Michael V Hyland: Alright. I love that you love that alright. And the last rhythm we're gonna cover today is sinus arrhythmia.

11 Mechanism of Sinus Arrhythmia

0:42:29 – 0:44:18



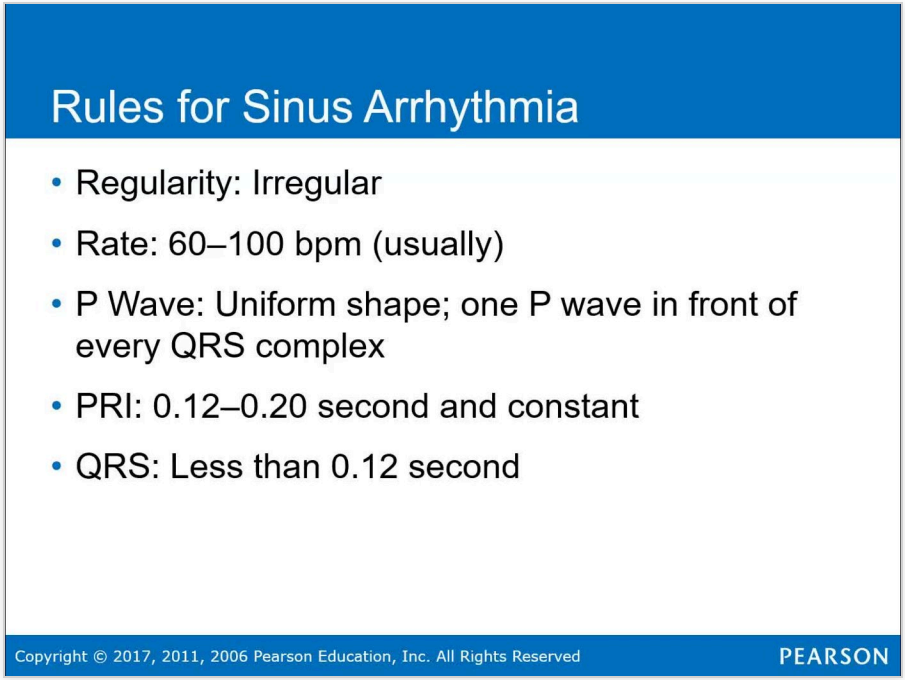
0:42:35 Michael V Hyland: So I just want to again focus a second of time here on the terminology. Michael V Hyland: The prefix A. Michael V Hyland: I know the English language is great, but prefix a in the English language means what without. Michael V Hyland: So when we say arrhythmia. Michael V Hyland: it means without rhythm. Michael V Hyland: and what what would that mean without rhythm? Okay, great. Michael V Hyland: So rhythm or rhythmic hold Michael V Hyland: would be even spacing

0:43:14 Michael V Hyland: without rhythm would be uneven, spacing. Okay. alright. So as a note in this rhythm is still the pacemaker. but the impulses are initiated in an irregular pattern, meaning R-r is irregular in this rhythm. Michael V Hyland: and what happens here is the rate changes with the patient's breathing pattern. The rate is going to increase Michael V Hyland: as the patient breathes in. So remember, we said, when the rate increases, our waves get closer together when the rate decreases, our waves get farther apart. So during inspiration. Michael V Hyland: the our waves are going to get closer together. That means the rate speeds up when they exhale, it's going to slow down. Our waves are going to stretch out.

0:44:05 Michael V Hyland: In any event. What's going to happen with this is R to R is going to be irregular. Michael V Hyland: Each impulse is conducted normally through to the ventricles meaning the Q. Rs is going to be narrowed.

12 Rules for Sinus Arrhythmia

0:44:18 – 0:45:53



0:44:18 Michael V Hyland: So let's look at our rules. This regularities is irregular. Michael V Hyland: The heart rate for this rhythm is usually between 60 to a hundred.

0:44:33 Michael V Hyland: I have seen this rhythm less than 60, Michael V Hyland: so if we identify this rhythm less than 60, Michael V Hyland: we got 2 problems. What are the 2 problems? Science free? It's bradycardic. And it's without rhythm.

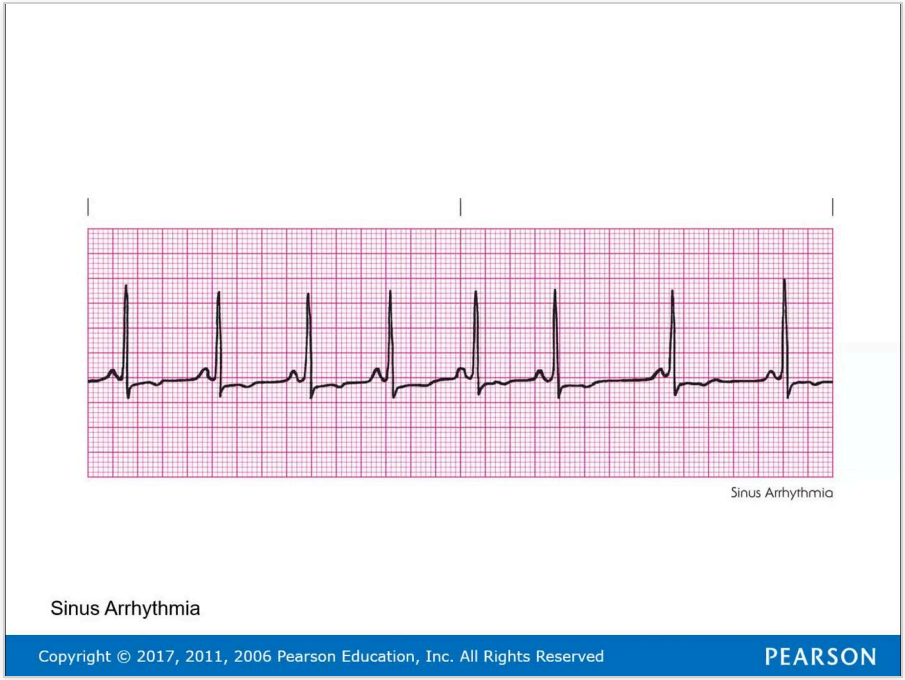
0:44:52 Michael V Hyland: So how would we interpret that rhythm? Michael V Hyland: Sinus Brady Arrhythmia? Michael V Hyland: Oh, didn't know you could combine 2 together. Interesting. Michael V Hyland: or if the heart rate was above 100, it might be sinus, tacky, arrhythmia or sinus tack, arrhythmia. Right? Michael V Hyland: Okay? So I might throw one or 2 of those at you. So beware. I've been known to sneak a couple things in. Michael V Hyland: I mean, my job is to challenge you, is it not? And make you learn that is my job. Michael V Hyland: I take this very seriously. Michael V Hyland: Alright. So the heart rate's usually 60 to a hundred.

0:45:34 Michael V Hyland: Okay, but it can be below 60 or rarely above a hundred. I'll give you a clue. I won't give you one above a hundred. Alright, I'll narrow it down for you.

0:45:46 Michael V Hyland: P. Waves will be upright in uniform. Michael V Hyland: Pr. I will be normal, and the Qrs will be narrow. So let's take a look at this. All right, let's count our spacing. Gonna strain your eyes one last time today.

13 Sinus Arrhythmia (ECG strip)

0:45:53 – 0:54:38



0:46:00 Michael V Hyland: Okay. Michael V Hyland: be okay. Michael V Hyland: I have a different way to describe, but I'll keep it to myself, for now. Lacey, note school excellent. I can tell you. My eye at that surgery.

0:46:35 Michael V Hyland: Your homework assignment tonight is stare at it for 4 h. Michael V Hyland: Exactly. We're ready. Michael V Hyland: What we get for the first spacing Michael V Hyland: cubicle.

0:48:39 Michael V Hyland: I like 19. Okay, because there's 2 halves on both ends. Michael V Hyland: Okay. Michael V Hyland: the first R wave is halfway in between a small box. The second R wave is more towards the middle. Michael V Hyland: So that's why I got 2 halves. I like 19. Michael V Hyland: What's the next? 1? 1818? What's the next 1? 60? I would call it 16 and a half.

0:49:15 Michael V Hyland: What about the next one. Becky? Let me Michael V Hyland: 17, I agree. What's the next one? It is 16. What's the next? One? 23 and a half? I agree, what's the next? 1, 22, and a half?

0:49:37 Michael V Hyland: So the spacing is not within the one and a half small box rule. Right? Michael V Hyland: So this rhythm is irregular. Michael V Hyland: Okay, how do we count the heart rate for near regular rhythm? Count the R's multiply by 10

0:49:57 Michael V Hyland: 80. II can curve. Michael V Hyland: he waves. Michael V Hyland: I believe they are upright. I think they look Michael V Hyland: pretty much the same. It's close enough that I would say, upright in uniform. Michael V Hyland: Pr. I.

0:50:29 Michael V Hyland: Or Michael V Hyland: and a half 3 and a half, 3 and a half. Michael V Hyland: Yeah, that's 3 and a half. Michael V Hyland: It's pretty consistent. 3 and a half. You want me come back there and help you

0:50:50 Michael V Hyland: want to comment on. Michael V Hyland: So even if we just look at this one here before the Qrs. And part is right there. and that memorial starts actually cleaning Michael V Hyland: 1, 2, 3, and a half Michael V Hyland: right? Michael V Hyland: And even this one alright. So this Pr, I Michael V Hyland: is point Michael V Hyland: 1 4. Michael V Hyland: Gotta get you used to Michael V Hyland: in time over there. Point 1 4. Alright. Let's measure the Qrs

0:51:52 Michael V Hyland: doing the same. Michael V Hyland: No. One. Michael V Hyland: Okay. Well, the phone player was maybe one. Michael V Hyland: All about.

0:52:28 Michael V Hyland: what'd you guys get was. Michael V Hyland: So I think this is kind of bordering somewhere between 2 and 2 and a half small boxes

0:52:41 Michael V Hyland: you want me to demonstrate. Michael V Hyland: Get in right now. Michael V Hyland: So. But if this Urs starts right there on that line, everybody sees that. Michael V Hyland: and it ends halfway in between right there.

0:52:58 Michael V Hyland: 1, 2 and a half. Michael V Hyland: Now, if I take this next one, it starts right there Michael V Hyland: right there, and it ends right there. So now I'm going to count backwards 1, 2, Michael V Hyland: and a half.

0:53:15 Michael V Hyland: I'm pretty confident this 2 and a half small boxes. Did you see what I did there? Okay. Michael V Hyland: so point 1 0. I should make you tell me what the time is? No, thank you. Michael V Hyland: Alright. So how do we interpret this rhythm? So it's irregular. But we have a normal heart rate. We have upright uniform. P waves. We have a normal pri and a normal Qrs. Michael V Hyland: so it's coming from the sinus. The only problem is, it's irregular. So that's why we would label it sinus arrhythmia Michael V Hyland: without rhythm. Michael V Hyland: Okay? Questions.

0:53:56 Michael V Hyland: comments. Everybody's had enough. Michael V Hyland: It's hard to do a lecture after the test. If you prepare so hard for the test, and then but I gotta cover what I need to cover in the time that Michael V Hyland: you need me to cover it. I know I apologize for that. I'm trying to say the forest. I told you that I'm gonna make the homework assignments like that.

0:54:30 Michael V Hyland: Let me stop the recording hold on Michael V Hyland: cool.